Suggested problems: 1-27 odd, 31, 33

Problem 1: Sketch the following vector fields in $\mathbb{R}^2$.

(a) $\vec{F}(x,y) = \langle 2, 3\rangle$ See class notes for sketch.
(b) $\vec{F}(x,y) = \langle y, 0\rangle$ See class notes for sketch.
(c) $\vec{F}(x,y) = \langle x, y\rangle$ See class notes for sketch.
(d) $\vec{F}(x,y) = \langle y, -x\rangle$ See class notes for sketch.
(e) $\vec{F}(x,y) = \dfrac{x\vec{i} + y\vec{j}}{\sqrt{x^2+y^2}}$ See class notes for sketch.

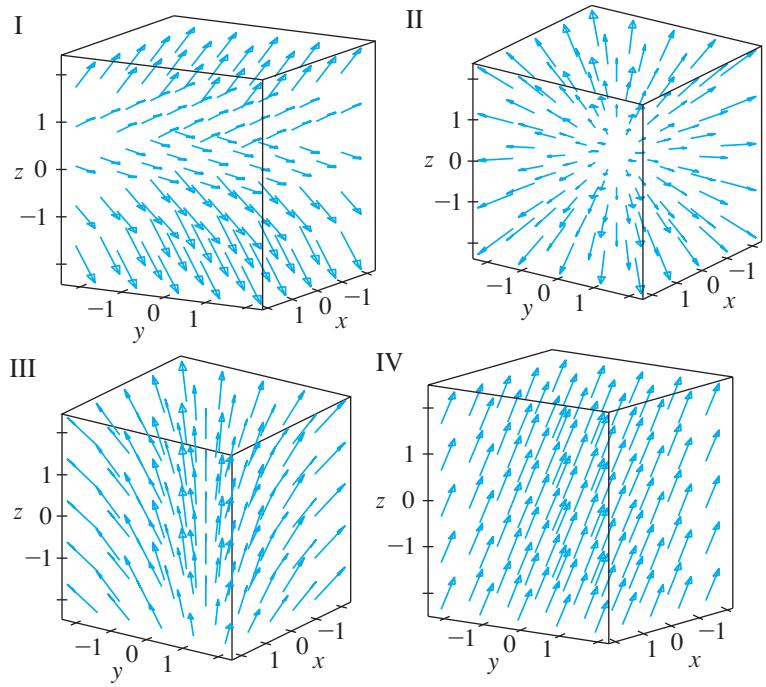
Problem 2: Match the vector fields in $\mathbb{R}^3$ to their formulas.

(a) $\vec{F}(x,y,z) = \langle 1, 2, 3\rangle$
IV

(b) $\vec{F}(x,y,z) = \langle 1, 2, z\rangle$
I

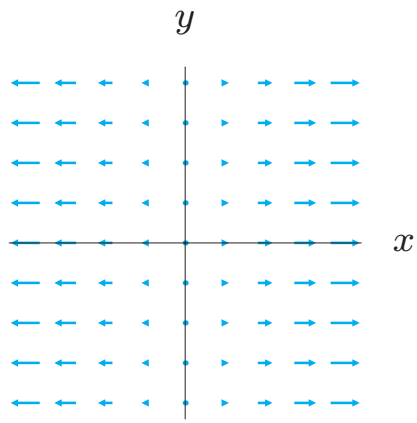
(c) $\vec{F}(x,y,z) = \langle x, y, 3\rangle$
III

(d) $\vec{F}(x,y,z) = \langle x, y, z\rangle$
II



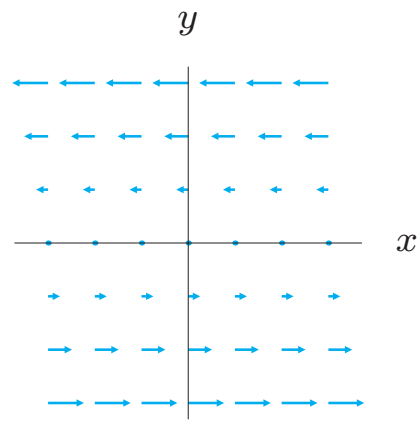
Problem 3: Find formulas for the vectors fields below.

(a)



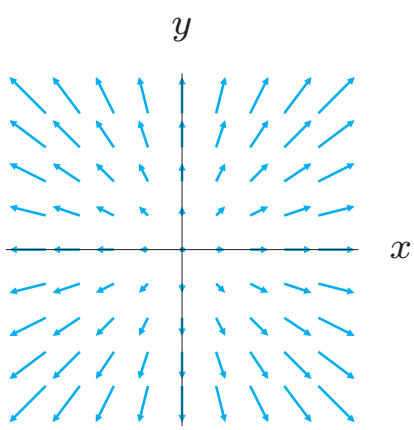
$\vec{F}(x,y) = \langle x, 0\rangle$

(b)



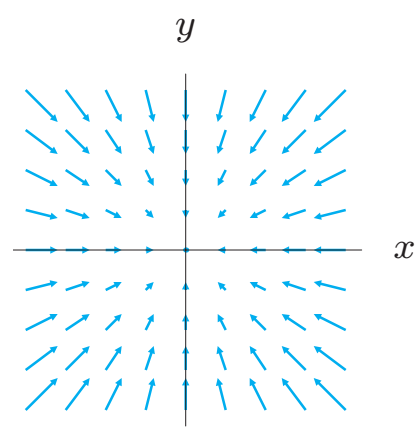
$\vec{F}(x,y) = \langle -y, 0\rangle$

(c)



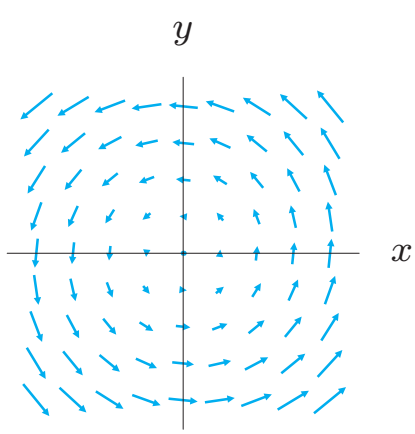
$\vec{F}(x,y) = \langle x, y\rangle$

(d)



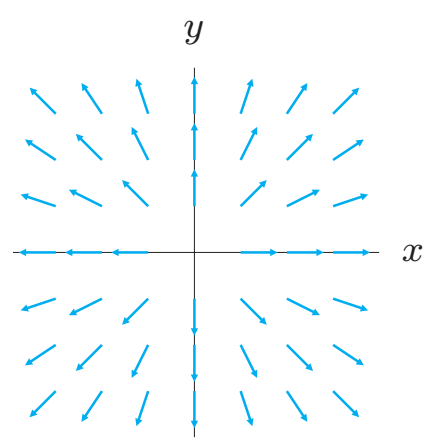
$\vec{F}(x,y) = -\langle x, y\rangle$

(d)



$\vec{F}(x,y) = \langle -y, x\rangle$

(f)



$\vec{F}(x,y) = \frac{\langle x,y\rangle}{\sqrt{x^2+y^2}}$

Problem 4: For each description of a vector field in parts (a)-(d), choose one or more of the vector fields (i)-(ix).

(a) Pointing radially outward, increasing in length away from the origin. iii

(b) Pointing in a circular direction around the origin, remaining the same length. ii

(c) Pointing toward the origin, increasing in length farther from the origin. iv

(d) Pointing clockwise around the origin. vi

(i) $\dfrac{\langle x, y\rangle}{\sqrt{x^2+y^2}}$

(ii) $\dfrac{\langle -y, x\rangle}{\sqrt{x^2+y^2}}$

(iii) $x\vec{i} + y\vec{j}$

(iv) $-x\vec{i} - y\vec{j}$

(v) $\langle -y, x\rangle$

(vi) $\langle y, -x\rangle$

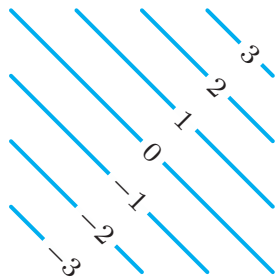
(vii) $\langle y, x\rangle$

(viii) $\dfrac{\langle x, y\rangle}{(x^2+y^2)^{3/2}}$

(ix) $-\dfrac{\langle x, y\rangle}{(x^2+y^2)^{3/2}}$

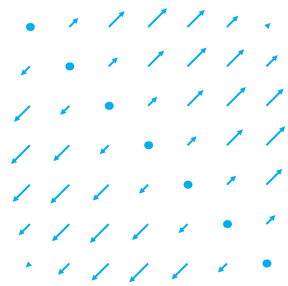
Problem 5: Match the contour plots in (a)-(d) with the gradient fields in (i)-(iv).

(a)

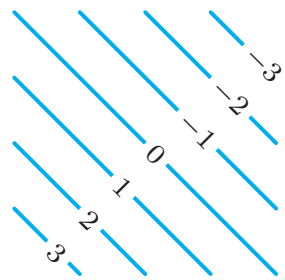


iii

(i)

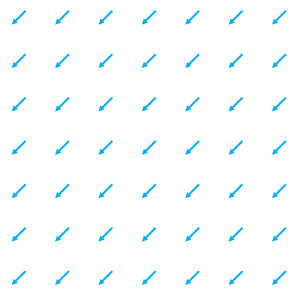


(b)

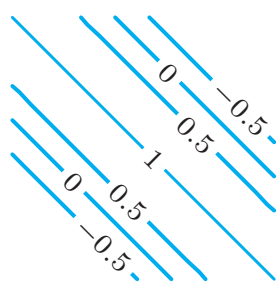


ii

(ii)

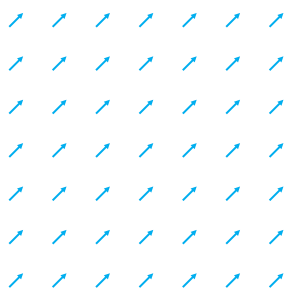


(c)

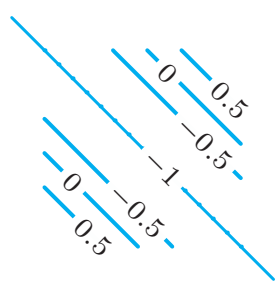


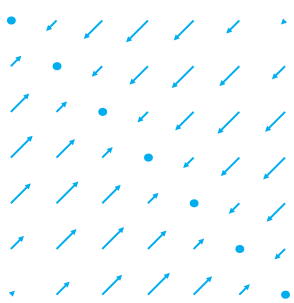
iv

(iii)



(d)



i

(iv)